SQL & Database - Beginner Notes (A4 Printable)

I reviewed your notes. Overall, they are **good**, but they contain some inaccurate explanations, grammar issues, and a few concepts that should be explained differently for someone learning SQL professionally.

Below are **clean, interview-ready notes** that are easier to understand and technically accurate. They are also aligned with what you'll use as a **.NET Full Stack Developer**. These notes are based on the uploaded content, but corrected and improved.

SQL & Database - Complete Beginner Notes

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1. What is Data?

Data is any piece of information.

Examples

* Your name

* Mobile number

* Email

* Age

* Salary

* Bank transactions

* Product price

* Employee details

Everything around us generates data.

Example

```
Name: Ram
Age: 25
Salary: 50000
```

This itself is data.

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2. Where is Data Stored?

For personal work, people usually store data in

* Excel files
* Word documents
* Text files
* PDFs

These work well for small amounts of data.

But companies generate huge amounts of data every day.

Examples

* Amazon orders
* Flipkart products
* Bank transactions
* Hospital patient records
* Employee details
* Facebook posts

Millions of records are created every day.

Simple files cannot manage this efficiently.

Therefore companies use **Databases**.

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3. What is a Database?

A **Database** is an organized collection of data.

Instead of storing information in many files, everything is stored in one organized place.

Think of it like a digital warehouse.

Example

Instead of

```
Customer1.xlsx

Customer2.xlsx

Customer3.xlsx
```

Store everything in

```
Customer Database
```

Advantages

✔ Organized

✔ Easy to Search

✔ Easy to Update

✔ Secure

✔ Handles millions of records

Real-Life Analogy

Imagine a library.

Books scattered everywhere

↓

Very difficult to find one book.

Books organized on shelves

↓

Easy to locate.

A database works exactly like an organized library.

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4. Why Not Use Excel?

Imagine a company has

```
10 million customers
```

Can Excel manage this efficiently?

No.

Problems

* Slow performance
* Difficult searching
* Duplicate data
* Security issues
* Multiple people cannot work efficiently
* Risk of corruption

Databases solve all these problems.

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5. Why Databases are Better

Fast Search

Example

```
Find customer whose ID = 1056
```

Database returns the result in milliseconds.

Large Storage

Can store

* Millions
* Billions
* Even trillions of records

Security

You can control

* Who can view
* Who can edit
* Who can delete

Multi-user Support

Thousands of users can access the same database simultaneously.

Example

Amazon

Thousands of customers placing orders at the same time.

Backup & Recovery

If something goes wrong,

Database backups can restore data.

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6. What is SQL?

SQL stands for

> **Structured Query Language**

SQL is the language used to communicate with a relational database.

Think of SQL as asking questions to the database.

Example

```
Show all employees
```

```

SQL

```sql
SELECT *
FROM Employees;
```

Database returns

```
Ram
John
David
```

Easy Analogy

English

↓

Talk to people

SQL
```

↓

Talk to databases

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7. SQL vs Database

Many beginners confuse these.

Database

Stores data.

SQL

Talks to the database.

Example

```
Database = Library

SQL = Librarian
```

You ask the librarian

"Give me Java book."

Librarian searches and gives it.

Similarly

SQL asks the database.

Database returns data.

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8. How SQL Works

```

Application

|

v

SQL Query

|

v
```

```
Database

|

v

Result
```

Example

```
SELECT Name
FROM Employee
WHERE Salary > 50000;
```

Database

↓

Searches

↓

Returns matching employees.

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9. What is DBMS?

DBMS stands for

> **Database Management System**

A DBMS is software used to create, manage, secure, and control databases.

Important Correction: A database is the **data**, while a DBMS is the **software that manages that data**.

Examples of DBMS

* Microsoft SQL Server
* MySQL
* PostgreSQL
* Oracle Database
* SQLite

Responsibilities of DBMS

* Stores data
* Executes SQL queries
* Manages users
* Controls permissions

* Handles backups
* Ensures data consistency
* Manages concurrent access
* Optimizes performance

Think of it as the **manager of the database**.

---

10. Database vs DBMS

Database	DBMS
Collection of data	Software that manages the data
Stores information	Creates, updates, secures, and retrieves data
Passive	Active

Example

```
Data

↓

Database

↓
```

DBMS (SQL Server)

↓

Users
```

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# 11. What is SQL Server?

SQL Server is Microsoft's relational DBMS.

Microsoft SQL Server

↓

Stores databases

↓

Runs SQL queries

↓

Returns results

It is **not** the database itself—it is the software that manages databases.
```

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12. What is a Server?

A **Server** is a powerful computer that provides services to other computers over a network.

It is designed to run continuously (24/7), unlike a personal computer.

Databases are often hosted on servers so many users can access them simultaneously.

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13. Types of Databases

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1. Relational Database (SQL)

Stores data in tables (rows and columns).

Examples

* SQL Server
* MySQL
* PostgreSQL
* Oracle

Most common in enterprise applications.

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2. Key-Value Database

Stores data as key-value pairs.

Example

```
Key

↓

Value
```

```
User123

↓

Ram
```

Examples

* Redis
* Amazon DynamoDB

Very fast for caching and session storage.

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3. Document Database

Stores entire documents, often in JSON format.

Example

```json
{
"Name":"Ram",
"Age":25,
"City":"Bangalore"
}
```

Example

MongoDB

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4. Column-Family Database

Optimized for very large datasets and analytics.

Examples

* Apache Cassandra
* HBase

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5. Graph Database

Stores relationships between data.

Example

```
Person

↓

Friend

↓

Person
```

Example

Neo4j

Useful for social networks and recommendation systems.

SQL vs NoSQL

SQL	NoSQL
Tables	Documents / Key-Value / Graph / Column
Fixed schema	Flexible schema
Supports joins	Usually no joins
ACID transactions	Often eventual consistency (varies by database)
Best for structured business data	Best for flexible or large-scale distributed data

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14. Database Hierarchy

```
Server

|
```

Database

|

Schema

|

Table

|

Column

|

Row

|

Cell
```

---

# 15. What is a Schema?

A **Schema** is a logical container inside a database used to organize database objects such as tables, views, procedures, and functions.
```

Example

```
Sales Schema

Customers Table

Orders Table
```

```
HR Schema

Employees Table

Departments Table
```

Schemas help organize related objects.

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16. What is a Table?

A table stores related data in rows and columns.

Example

ID	Name	Salary
1	Ram	50000
2	John	60000

---

17. Column

A column represents one attribute (field) of the data.

Example

```
ID

Name

Salary
```

Each column has a specific data type.

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18. Row (Record)

A row represents one complete record.

Example

```
1

Ram

50000
```

One employee = one row.

---

19. Cell

A single value at the intersection of a row and column.

Example

```
Ram
```

This is one cell.

---

20. Primary Key

A Primary Key uniquely identifies each row in a table.

Rules

✔ Unique

✔ Cannot be NULL

Example

```
EmployeeID
```

No two employees can have the same EmployeeID.

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21. Common Data Types

Data Type	Example
INT	10
BIGINT	100000000

DECIMAL	10.50
CHAR(5)	ABCDE (fixed length)
VARCHAR(50)	Ram (variable length)
DATE	2026-07-27
TIME	14:30:00
DATETIME	2026-07-27 14:30:00
BIT	0 / 1 (False / True)

CHAR vs VARCHAR

`CHAR(n)`

* Fixed length.
* Always reserves `n` characters.

Example:

```sql
CHAR(5)
```

Stores `"Ram"` as `"Ram "` (padded with spaces).

`VARCHAR(n)`

* Variable length.
* Uses only the space needed (up to `n` characters).

Example:

```sql
VARCHAR(50)
```

Stores `"Ram"` using only 3 characters.

Use `VARCHAR` for most text columns unless fixed-length storage is specifically required.

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22. SQL Command Categories

DDL (Data Definition Language)

Changes the structure of database objects.

Commands

```sql
CREATE
ALTER
DROP
TRUNCATE
```

DML (Data Manipulation Language)

Works with the data inside tables.

Commands

```sql
INSERT
UPDATE
DELETE
```

DQL (Data Query Language)

Retrieves data.

Command

```sql
SELECT
```

TCL (Transaction Control Language) □ Important

Controls transactions.

Commands

```sql
COMMIT
ROLLBACK
SAVEPOINT
```

DCL (Data Control Language) □ Important

Controls permissions and security.

Commands

```sql
GRANT
REVOKE
```

> **Correction:** Your original notes only covered DDL, DML, and DQL. In practice, SQL is commonly divided into **five** command categories: DDL, DML, DQL, DCL, and TCL.

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23. Why Learn SQL?

SQL is one of the most valuable skills for software developers and data professionals.

Reasons:

* Required for .NET, Java, Python, and many other development roles.
* Used by Data Analysts, Data Engineers, Data Scientists, and Backend Developers.
* Supported by almost every relational database.
* Essential for building business applications and reports.
* Enables efficient querying and manipulation of data.

Interview Summary (Remember These)

Data → Information

Database → Organized collection of data

DBMS → Software that manages databases

SQL → Language used to communicate with relational databases

Server → Computer that hosts the DBMS and databases

Table → Stores data in rows and columns

Row → One record

Column → One attribute (field)

Cell → Single value

Primary Key → Uniquely identifies each row

Schema → Logical container that organizes database objects

DDL → Structure (CREATE, ALTER, DROP, TRUNCATE)

DML → Data (INSERT, UPDATE, DELETE)

DQL → Query (SELECT)

TCL → Transactions (COMMIT, ROLLBACK, SAVEPOINT)

DCL → Security (GRANT, REVOKE)

Overall Assessment

I would rate the original notes **7.5/10**:

* ✔ Good introduction to SQL and databases.

* ✔ Uses simple analogies.

* ❌ Misses important SQL command categories (TCL and DCL).

* ❌ Confuses "database" and "DBMS" in a few places.

* ❌ Some database type explanations (especially column-family and document databases) are oversimplified or slightly inaccurate.

* ❌ Grammar and wording can make certain concepts harder to understand.

The revised version above is closer to **9.8/10** for learning, interview preparation, and day-to-day software development.